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The Use of Video Modeling via a Video iPod and a System of Least Prompts to Improve Transitional Behaviors for Students with Autism Spectrum Disorders in the General Education Classroom

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This study evaluated the efficacy of video modeling delivered via a handheld device (video iPod) and the use of the system of least prompts to assist elementary-age students with transitioning between locations and activities within the school. Four students with autism learned to manipulate a handheld device to watch video models. An ABAB withdrawal design was used to evaluate a functional relation between video modeling and the percentage of independent transitions by the student. The data indicate that all participants began transitioning more independently after the intervention was introduced and that their performance decreased with withdrawal of intervention. These results are presented in the context of how a portable video delivery system can aid students who spend considerable portions of their day in general education settings where traditional means of delivering video models (e.g., televisions and computers) may not be as accessible.

Keywords: *autism spectrum disorders; video modeling; intervention; behavior; inclusion; elementary*

Children with autism spectrum disorders (ASD) often struggle in situations involving change in the environment (American Psychiatric Association, 2000). These difficulties with environmental change often manifest in severe behavior problems during times of transition. Behavior problems, including heightened aggressiveness, increased anxiety, becoming distressed, and refusing to participate in the transition, may be observed in individuals with ASD. With the increase of educating students with disabilities in inclusive settings, students with ASD will be expected to transition more often in school settings. In an elementary school setting, students are expected to walk appropriately in the hallway to transition to and from exploratory classes (e.g., music, art, and physical education), the cafeteria, playground, bathroom, and buses. With that in mind, an important goal of educators is to identify and implement the use of positive behavioral supports to increase successful environmental transitions and reduce problem behaviors associated with transitional situations.

Recent favorable treatments for students with ASD have focused on the use of video modeling. Video modeling involves a student watching another person engaging in target behaviors or skills on a video and then performing the behaviors or skills. A specific application of video modeling is video self-modeling, in which the target student serves as the model and views himself or herself performing the target behaviors accurately and independently

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(Dowrick, 1999). Moreover, video self-modeling often incorporates the person's point of view, that is, by "recording tapes using the perspective of the person who is the target of the intervention and showing him or her the target behavior from the viewer's vantage point but without showing the entire person who is modeling the behavior" (Hine & Wolery, 2006, p. 84). Researchers have posited that visual supports are especially effective for students with autism because their visual perception abilities are often seen as an area of strength (Quill, 1997). Also, individuals with autism are known to favor visual treatment approaches (e.g., MacDuff, Krantz, & McClannahan, 1993; Pierce & Schreibman, 1994). Hermelin and O'Connor (1970) reported that children with autism encode visuospatial information better than auditory information. Moreover, children with autism, with both higher and lower IQ scores, performed better on visuospatial tasks compared to tasks that required social or language reasoning (Lincoln, Courchesne, Kilman, Elmasian, & Allen, 1988). In addition, visual supports were used successfully for children with autism to maintain attention, understand spoken language, and sequence and organize their environment (Hodgdon, 1995). When instruction was facilitated with visual cues, individuals with autism acquired, maintained, and generalized a range of academic (e.g., MacDuff et al., 1993), functional (e.g., Vaughn & Horner, 1995), domestic or self-care (e.g., Pierce & Schreibman, 1994), leisure (e.g., D'Ateno, Mangiapanello, & Taylor, 2003), and communication skills (e.g., Bondy & Frost, 1994; Krantz & McClannahan, 1998).

Video-based instruction also was suggested as a viable teaching tool for students with autism because students view positive and successful behaviors as opposed to negative and unsuccessful behaviors (Dowrick, 1999). The use of video modeling is rooted in Bandura's (1977) social learning theory, a basic learning mechanism that has broad applicability to teaching new skills. This theory states that individuals can acquire a behavior by watching a model rather than through direct personal experience. A model is the person demonstrating the behavior. The observer acquires information from a model through watching and attending to the model. From this symbolic representation, a learner can retrieve information at any time and use it to guide behavior.

Several literature reviews (Ayres & Langone, 2005; Delano, 2007; Hitchcock, Dowrick, & Prater, 2003; Mechling, Pridgen, & Cronin, 2005) and a meta-analysis (Bellini & Akullian, 2007) have demonstrated the effectiveness regarding the use of video-modeling procedures. Video modeling was used successfully to teach students with ASD communication (e.g., Taylor, Levin, & Jasper, 1999), conversational (e.g., Sherer et al., 2001), play (e.g., Nikopoulos & Keenan, 2004); purchasing (Haring, Kennedy, Adams, & Pitts-Conway, 1987), community (e.g., Cihak, Alberto, Taber-Doughty, & Gama, 2006), domestic (e.g., Lasater & Brady, 1995); and vocational (e.g., Alberto, Cihak, & Gama, 2005) skills. In the Bellini and Akullian (2007) meta-analysis of video modeling, the results indicated moderate intervention effectiveness for targeting social-communication and functional skills. The results also indicated moderate effects for both maintenance and generalization.

However, video-modeling studies targeting behavioral functioning have been limited. Only two studies were found that targeted specifically the reduction of problem behaviors (Buggey, 2005; Schreibman, Whalen, & Stahmer, 2000), and two studies targeted off-task behaviors (Coyle & Cole, 2004; Hagiwara & Myles, 1999). Buggey (2005) examined the effects of video modeling for two students with ASD who engaged in tantrum or aggressive behaviors (i.e., pushing others). Videos included filming the student performing positive behaviors under conditions that usually triggered problem behaviors. The results indicated that the video-modeling intervention successfully reduced the duration of tantrums and the occurrence of aggressive behaviors. Moreover, students' problem behaviors maintained at low levels 3 weeks later. Similarly, Schreibman and colleagues (2000) used video priming as a means for reducing tantrum behaviors of three children with autism during transition situations. The children often displayed disruptive behaviors when transitioning from their home to different stores in the community. Videos included a videotaped recording, from the child's point of view, of the setting where the child would be transitioning. Similar to Buggey's study, the child watched the video just prior to the transition. However, unlike Buggey's study, no models appeared in the videos. The results indicated that the number of tantrum occurrences decreased during transitional situations following the use of video priming. Tantrum behaviors also maintained at low levels for 4 weeks during follow-up probes.

Several mediums were used successfully, as delivery systems, to show students the videos, including (a) television and videocassette recorder, (b) computer, and (c) handheld device (e.g., personal digital assistant [PDA], video iPod). Most of the video-modeling research studies incorporated the use of a television with a videotape recorder (Mechling et al., 2005). For example, Coyle and Cole (2004) used a television and videocassette recorder to show three elementary students with autism videos of themselves "working very well" (p. 8). Students watched the videos in a small therapy room near the student's

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classrooms each morning. The results indicated considerable decreases in off-task behavior following the video-modeling intervention.

As technological gains are made and accessibility to these technologies is becoming more readily available, recent video-modeling research has begun to explore different mediums. Cihak and Schrader (2008) used a laptop computer to show adolescent students with autism video clips of themselves (video self-modeling) independently and correctly performing vocational tasks. Because the video clips were delivered via a laptop computer, the students were able to watch the performance of the task in the setting where they were working and were ultimately expected to perform the target skill. In addition, if the student required additional assistance to complete the task, he or she simply replayed the video. Ayres and Langone (2005) suggested that video modeling was particularly useful when using digital video provided by a computer, because digital mediums can be easily integrated into other computer programs.

A third medium of video modeling used was the handheld computer (e.g., PDA). Cihak, Kessler, and Alberto (2008) examined the efficacy of video modeling using a commercially produced handheld computer to facilitate independent transitions during vocational training for students with severe disabilities. Students used the handheld computer to watch videos clips of themselves initiating and completing a task and transitioning to their next task located in a different place. The results indicated that the use of handheld computers to display video clips was a viable alternative medium. Video clips can be repeated easily and as often as necessary, various skills and behaviors can be made available, and the videos can be used with many students. In addition, the small size of the handheld device and its portability were the most convincing advantages as well as taking up a small amount of storage space, allowing students to take the device with them anywhere.

The videotaping and viewing of videos can occur in a wide variety of settings, including the school, home, or community settings (Delano, 2007). However, Dowrick (1999) review of the video-modeling literature noted that most of the studies occurred in nonschool settings, such as (a) clinical settings, (b) residential hospitals, (c) day treatment facilities, (d) home environments, and (e) community settings. Moreover, most of school-based research occurred in the special education classroom. Researchers have suggested that video modeling could be used as an effective tool to enhance behavioral skills and to reduce the potential risk of school failure in the general education classroom (Dowrick, 1999). However, few studies have focused on behavioral functioning in the general education classroom for students with ASD.

Educational experiences in the general education classroom provide students with disabilities the opportunity to learn and to practice skills they need to fully participate in inclusive environments. Harrower (1999) reported positive outcomes in terms of attitudes, acceptance, knowledge of disabilities, friendships, and reduction in negative stigma when students with disabilities were taught in inclusive classrooms. Rafferty, Piscitelli, and Boettcher (2003) also noted positive gains in language development and social competency when students with severe disabilities were included in general education classrooms. Additionally, there is empirical evidence that suggested some groups of students with disabilities who were placed in general educational classrooms do no worse academically than if they were placed outside the general educational classrooms (Farrell, 2000; Lindsay, 2007; Lunt & Norwich, 1999).

The purpose of this study was to examine the effects of using video-modeling procedures by means of a video iPod and the use of the system of least prompts in a general education classroom for students with autism. Specifically, what were the effects of using video modeling via a video iPod and the system of least prompts to increase independent transitions from place to place for elementary students with autism? Also, what was the social validity or acceptability of using these intervention procedures to facilitate independent transitions in the general education setting?

Method

Participants

Four elementary students diagnosed with autism participated in this study. Students were diagnosed by their private physician. Also, all students were evaluated by their local school system's school psychologist and determined to be eligible for special educational services under the autism category. Each was selected to participate based on (a) an Individualized Education Program objective related to improving adaptive social behavior skills, (b) difficulty with transitions, (c) no hearing or vision impairments that might impede video instruction, (d) agreeing to participate in the study, and (e) parental permission.

Table 1 displays autism assessment scores for all students. All assessments were conducted within the past 2 years of this study. Ian was a 6-year-old first-grade student. The administration of the *Childhood Autism Rating Scale* (CARS; Schopler, Reichler, & Renner, 1988)

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Table 1
Students' Autism Assessment Scores

Student	Instrument/Subtest	Score	Standard Score	Percentile	Interpretation
Ian	CARS	45	—	—	Severely autistic
Jose	GARS-2				Very likely of autism
	Stereotype behaviors	36	16	98	
	Communication	39	18	>99	
	Social interaction	38	16	98	
	Autism index	—	143	>99	
Ida	GARS-2				Very likely of autism
	Stereotype behaviors	28	13	84	
	Communication	32	15	95	
	Social interaction	29	12	75	
	Autism index	—	122	93	
Dave	GARS-2				Very likely of autism
	Stereotype behaviors	23	11	63	
	Communication	19	10	50	
	Social interaction	17	8	25	
	Autism index	—	98	45	

Note: CARS = *Childhood Autism Rating Scale* (Schopler, Reichler, & Renner, 1988); GARS-2 = *Gilliam Autism Rating Scale–Second Edition* (Gilliam, 2006).

placed Ian within the severe autistic range. The *Developmental Profile–II* (DP-II; Alpern, Boll, & Shearer, 1988) indicated an IQ equivalence score of 45, suggesting moderate intellectual disability. Jose was a 7-year-old second-grade student. The administration of the *Gilliam Autism Rating Scale–Second Edition* (GARS-2; Gilliam, 2006) placed Jose within the severe autistic range, and the *Wechsler Intelligence Scale for Children–IV* (WISC-IV; Wechsler, 2004) indicated an IQ equivalence score of 50, suggesting moderate intellectual disability. Ida also was a 7-year-old second-grade student. The GARS-2 placed Ida within the severe autistic range and the DP-II indicated an IQ equivalence score of 48, suggesting moderate intellectual disability. Dave was an 8-year-old third-grade student. The administration of the GARS-2 placed Dave within the severe autistic range. The WISC-IV indicated an IQ equivalence score of 35, suggesting severe intellectual disability. All students communicated primarily using gestures, pictures, and single-word signing. Although spontaneous communication was rarely observed, students were observed to reliably answer yes-or-no questions. Verbal speech was seldom observed, which was usually echolalic or unintelligible in nature.

In addition, four general education teachers and four paraprofessionals also participated. All teachers were certified in K-6 elementary education and had taught for a minimum of 3 years. Paraprofessionals had worked at their respective schools for a minimum of 2 years. Although the paraprofessionals worked extensively with the participating students in the classroom, they also assisted other students with disabilities in the classroom, too. That is, paraprofessionals were assigned and expected to work with all students in the classroom instead of specifically working with the students with autism in a one-to-one situation. Paraprofessionals worked under the direction of both a general education teacher and a special education teacher.

Setting

Four elementary schools served as the location of this study. Each participating student attended a local neighborhood school. Pretraining instruction using the video-modeling procedures by means of a video iPod occurred in each student's general education classroom. Each classroom was staffed by a certified elementary teacher and paraprofessional. Baseline, pretraining, the handheld video-modeling intervention, and maintenance phases occurred in the general education setting, because students were fully included in all general education classes.

Materials

All videos were created using a Sony 72X Digital Zoom camera and iMovie 4.01 software for the iMac G4. Ten videos for each student were developed showing a positive self-model (Dowrick, 1999) of the student transitioning independently from place to place. A combination of role-playing and videotaping the student enough times until he or she transitioned independently was used to

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Table 2
Students' Transitions

Transition	Student			
	Ian	Jose	Ida	Dave
1	Bus to classroom	Bus to classroom	Bus to classroom	Bus to classroom
2	Classroom to music room	Classroom to gymnasium	Classroom to bathroom	Classroom to cafeteria
3	Music room to bathroom	Gymnasium to bathroom	Bathroom to classroom	Cafeteria to bathroom
4	Bathroom to classroom	Bathroom to classroom	Classroom to cafeteria	Bathroom to classroom
5	Classroom to cafeteria	Classroom to cafeteria	Cafeteria to bathroom	Classroom to music room
6	Cafeteria to bathroom	Cafeteria to bathroom	Bathroom to classroom	Music room to classroom
7	Bathroom to classroom	Bathroom to classroom	Classroom to art room	Classroom to playground
8	Classroom to playground	Classroom to playground	Art room to playground	Playground to bathroom
9	Playground to classroom	Playground to classroom	Playground to classroom	Bathroom to classroom
10	Classroom to bus	Classroom to bus	Classroom to bus	Classroom to bus

record each student's performance. Students' target behaviors were edited out, and only the desired performance was presented. A blend of video self-modeling and personal point-of-view modeling (Hine & Wolery, 2006) was used to show each student's transitioning. Each video clip showed the student, as the model, lining up with his or her classmates, walking in the hallway, and entering the next location. A personal point-of-view perspective also was included to show the students' vantage point as they walked to their destination. The length of video clips ranged from 2 to 5 min depending on the distance of the designated location.

After each video was edited appropriately, videos were downloaded into a 30 GB Apple video iPod to view the videos. The iPod had a height of 4.1 inches and width of 2.4 inches, was 0.5 inches thick, and weighed 4.8 ounces. Each was encased in an iSkin silicone protective case with a clear plastic overlay that allowed for ease in viewing and operation. The back of the case contained a plastic clip that allowed students to secure the iPod to a belt or pocket if they chose.

Response Measurement and Data Collection

All students had difficulty transitioning independently from place to place. Event recording was used to record the number of times a student independently transitioned or required assistance to transition to a different location in the school. Teacher assistance was required when student target behaviors were observed or the student did not physically move independently to the next location and required adult assistance. Each student's teacher reported target inappropriate behavior. The problem behavior that Ian's teacher was most concerned with was physical aggression. Physical aggression was defined as pinching and slapping another person when walking in the hallway during transition times. Jose's teacher also was concerned with physically aggressive behaviors. Jose's physical aggression was defined as pulling others' hair or slapping another person in the hallway. Ida's teacher was concerned with Ida's running away from the group (elopement). Elopement was defined as moving at least 10 feet away from her classmates in the hallway. For Dave, the target behavior was sitting on the floor. Sitting on the floor was defined as Dave's body on the hallway floor with legs crossed.

The appropriate alternative behavior for all students was transitioning independently from place to place. An independent transition was defined as walking in the hallway with classmates from place to place without engaging in the targeted inappropriate behavior. A student's independent transition was recorded as correct when the student lined up with his or her classmates, physically moved to the new location, and completed the first step of the required task in the new location. A transition was considered assisted if the student required adult assistance contingent on the performance of targeted inappropriate behaviors at any time after being prompted to line up until performing the first step of a task in the new location or if the student did not physically move independently to the next location and required adult assistance.

Ten daily transitions from place to place were identified for each student. Table 2 lists each student's transition. These transitions encompassed all routine transitions that all students in the classroom experienced throughout the school day. All transitions also followed the same order daily. The total number of independent transitions was divided by 10 to compute a percentage of independent transitions for each student.

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Experimental Design

An ABAB design (Barlow & Hersen, 1984) was used to determine the efficacy of video-modeling procedures using a video iPod and student independent transitions. The ABAB design allowed sequential application, comparison effectiveness, and intrasubject replication of effect. The study included five phases: baseline, handheld video-modeling procedures, no handheld video-modeling procedures, handheld video-modeling procedures reinstated, and maintenance phases. Prior to baseline, a pretraining period occurred.

Experimental Procedures

Pretraining. Prior to baseline, students participated in a pretraining period to ensure that students could attend to and perform the directions displayed in a video format. For the first phase of pretraining, students were instructed on how to operate the handheld video device. They were instructed to physically turn on the device and to select a color-cued hardware button to play the video. A model-lead-test procedure (Adams & Engelmann, 1996) was used to instruct the students on the use of the handheld device. The teacher modeled the necessary steps for accessing the video clips, led the students as they accessed and watched the video, and tested the student's performance of using the handheld device and accessing the videos independently.

In the second phase, students were required to turn on the device, play the video, watch the video, and follow each instruction of the video. A prerecorded video of the student performing a familiar two-step task normally associated with the student's morning classroom routine was used. Students watched themselves either closing the door and hanging up their coat or sitting down at their desk and picking up a pencil. If the students activated and watched the video as well as completed the two-step task correctly, the student received verbal praise from the teacher. If an incorrect step (error) was performed, the teacher said, "[Name], let's watch the video again" and prompted the student to turn on the device, watch the video, and follow each instruction of the video. If incorrect steps continued, a least-to-most prompt hierarchy was used to complete the individual steps of the task until the student performed the correct response without assistance. A 4-s interval between prompts was implemented. The least-to-most prompt hierarchy consisted of the following levels across the instructional session: (a) verbal prompt (e.g., "Do you see yourself [e.g., closing the door, picking up a pencil]?"), (b) gesture plus verbal explanation (e.g., pointing to the video stimulus and providing verbal explanation), (c) modeling plus verbal explanation (e.g., teacher performing the step and providing verbal explanation), and (d) physical assistance plus verbal explanation (e.g., holding the student's hand, guiding the correct response, and providing verbal explanation). Each student was required to reach a criterion of 100% accuracy for three consecutive sessions.

Baseline. During baseline, the number of independent transitions from place to place was recorded. If a student was unable to independently move to the next location or performed a targeted behavior, the transition was recorded as *assisted*. Assistance for all transitions was provided by an external source only after the demonstration of the student's target behavior. Data were collected until a stable baseline was achieved for a minimum of five sessions.

Handheld video-modeling procedures. All students started their school day arriving on the school bus. Prior to each transition, students were provided with the handheld video device and instructed to turn it on. After the device was activated, the student pressed the button to play the video. Students watched the corresponding video of themselves walking to the designated location. After the student watched the video of themselves transitioning from place to place independently, the teacher prompted the student to line up. Students received verbal praise contingent on transitioning independently from place to place. Similar to pretraining, if the student demonstrated a target behavior and required assistance in transitioning, the teacher said, "[Name], let's watch the video again" and prompted the student to turn on the device and watch the video. If target behaviors continued, a least-to-most prompt hierarchy, similar to the pretraining phase, was used to complete the transitions until the student performed the correct response without assistance. This process continued until all 10 transitions were completed. Each student was required to reach a criterion of 100% independent transitions for three consecutive sessions.

No handheld video-modeling procedures. The criteria for intervention withdrawal occurred when the data indicated that the student transitioned independently 100% of the time for three consecutive sessions. Similar to the baseline phase, the number of independent transitions from place to place was recorded for each student. Data were collected for a minimum of three sessions or when the mean of this phase returned to performances similar to the baseline phase.

Handheld video-modeling reinstated. The criteria to reinstate the intervention phase occurred when the mean

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of the intervention withdrawal phase returned to within close proximity of the mean of the baseline phase or a trend occurred in the opposite therapeutic direction of the intervention. Similar to the previous handheld modeling phase, prior to each transition, students were provided with the handheld video device and were instructed to turn on the device. Students then pressed the button to play and watch the video of themselves walking to the designated location. After the students watched the video, the teacher prompted the student to line up. Students continued to view the video clips prior to transitioning from place to place daily until the maintenance probes.

Maintenance procedures. Follow-up probes were collected 9 weeks after the student met the acquisition criterion. Similar to baseline conditions, students did not view the video prior to transitional situations. Follow-up probes occurred in the settings where the student was initially trained. Follow-up probes were collected to determine if the initial instruction affected the student's performance over time.

Reliability

Interobserver reliability data were collected simultaneously by a member of the research team (fourth author) and the classroom teachers. Interobserver data were collected during 60% of the sessions of baseline and during 33% of the sessions in each subsequent phase for all students. Observers independently and simultaneously recorded the number of students' independent or assisted transitions from place to place. Interobserver agreement was calculated by dividing the number of agreements of students' responses by the number of agreements plus disagreements and multiplying by 100. Interobserver reliability ranged from 98% to 100%, with a mean of 99% agreement. The mean interobserver reliability agreement for each student across conditions was Ian, 100% (kappa = 1.00); Jose, 98% (kappa = .93); Ida, 100% (kappa = 1.00); and Dave, 99% (kappa = .94).

Procedural reliability data were collected by a member of the research team (fourth author) during 33% of intervention phases for all students. Procedural integrity measures verified the teachers' performance of providing the handheld video device to the students, prompting the students to activate the device, and asking the students to line up. In addition, measures confirmed that the teachers delivered verbal praise contingent on independent transitions or redelivered the handheld video device, prompted the student to activate the device, and implemented a 4-s, least-to-most prompting hierarchy contingent on the performance of the student's target behavior. The teachers were trained using an itemized checklist that listed each step to be performed. The teachers were considered successfully trained after completing 100% of the checklist for three consecutive sessions. The procedural agreement level was calculated by dividing the number of observed teachers' behaviors by the number of planned teachers' behaviors and multiplying by 100 (Billingsley, White, & Munson, 1980). Procedural reliability ranged from 99% to 100%, with a mean of 99%. The mean procedural reliability agreement for each student across conditions was 99% for Ian (kappa = .94) and 100% (kappa = 1.00) for Jose, Ida, and Dave.

Social Validity

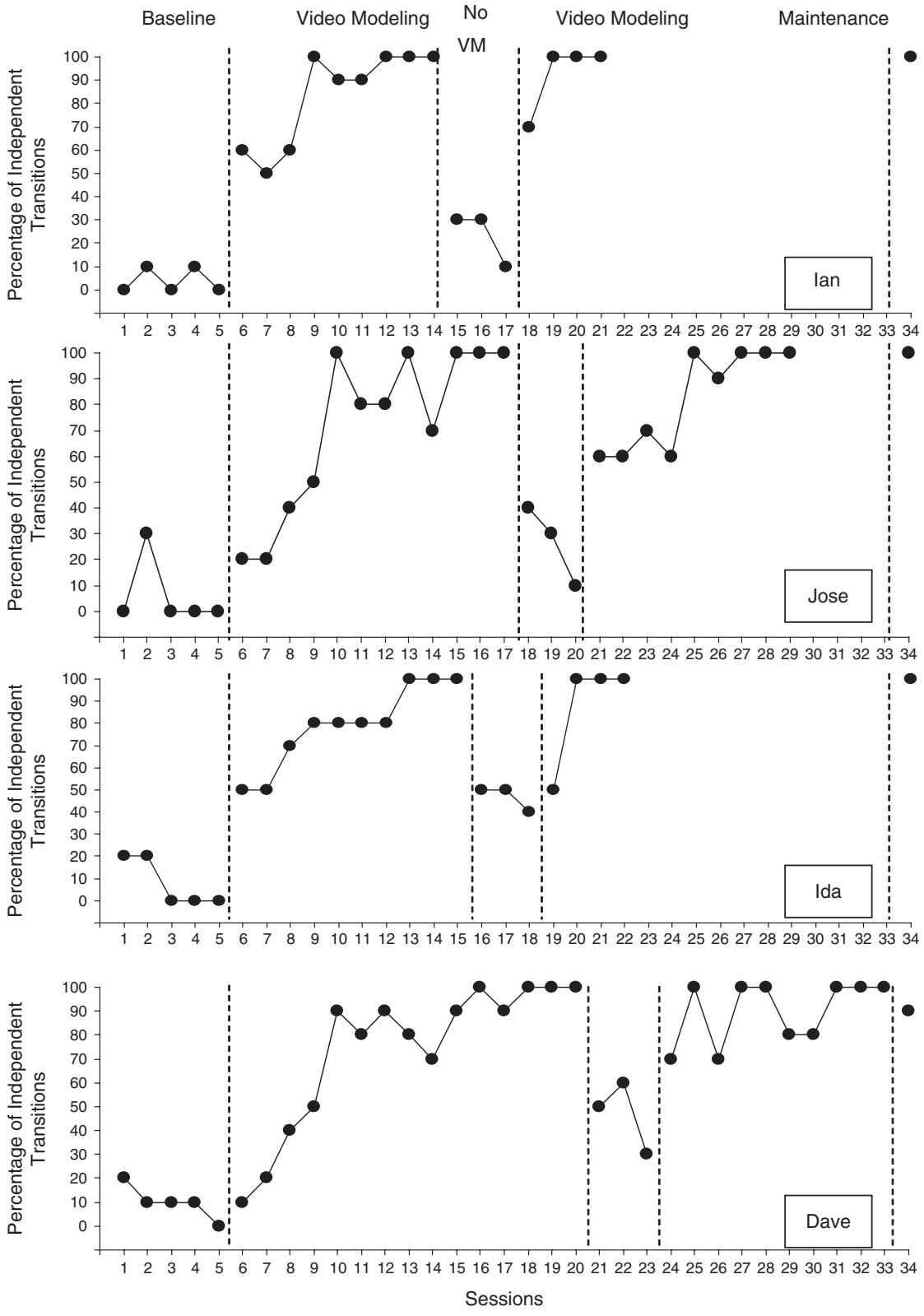
Following the conclusion of the study, the four participating teachers completed the *Intervention Rating Profile–15* (IRP-15; Martens, Witt, Elliot, & Darveaux, 1985) to assess the social validity and acceptability of using video-modeling procedures by means of a video iPod to increase independent transitions from place to place for elementary students with autism. The IRP-15 is a 15-item Likert-type scale that assesses general acceptability of interventions. The Likert-type scale ranges from 1 (*strongly disagree*) to 6 (*strongly agree*). Scores generated by the IRP-15 range from 15 to 90. Higher scores indicate better acceptance of interventions, and ratings higher than 52.5 were considered to reflect acceptability by the rater (VonBrock & Elliott, 1987). In addition, the participating students were asked, (a) "Did you like watching the videos of yourself?" (b) "Did the videos help your walking in the hallway?" (c) "Did the videos help you know what you have to do?" (d) "What did you like about the videos?" (e) "Would you like to watch more videos of yourself learning new things?" and (f) "Would you tell other kids about the videos?"

Results

The number of independent transitions made by students during baseline, handheld video modeling, no handheld modeling, and maintenance phases is presented in Figure 1. Overall, students demonstrated limited ability to transition independently from place to place during baseline. Overall, the mean number of independent transitions made by students during baseline was 7%. When students used the handheld video device during intervention, ascending trends were observed and the mean percentage of independent transitions increased to 77%. When the video-modeling intervention was withdrawn, descending trends were observed and the mean

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Figure 1
Students' Percentage of Independent Transitions With and Without the Use of the Handheld Video-Modeling Intervention



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percentage of independent of transition decreased to 36%. However, the mean percentage of independent transitions increased to 88% and ascending trends were observed when the handheld video-modeling intervention was reimplemented. Moreover, students maintained a mean level of 98% independent transitions 9 weeks later.

Ian demonstrated a mean of 4% independent transitions from place to place during baseline. During the handheld video-modeling phase, Ian increased independent transitions to a mean of 83% (range = 50% to 100%). Video clips were replayed as an error correction procedure during a total of 15 transitions (i.e., Sessions 6, 7, 8, 10, and 11). Of the 15 transitions, 6 transitions included only a replay of the video, 7 transitions included an additional verbal prompt, and 2 transitions included both verbal and gestural prompts. When the video-modeling intervention was withdrawn, Ian's independent transitions decreased to a mean of 23% (range = 10% to 30%). However, when the handheld video-modeling intervention was reimplemented, Ian's independent transitions from place to place increased to a mean of 93% (range = 70% to 100%). Video clips were replayed during 3 transitions (i.e., Session 18), and Ian required an additional verbal prompt for one of the transitions. Ian maintained a level of 100% independent transitions 9 weeks later.

Jose demonstrated a mean of 6% independent transitions from place to place during baseline. During the handheld video-modeling phase, Jose increased independent transitions to a mean of 72% (range = 20% to 100%). Video clips were replayed during a total of 34 transitions (i.e., Sessions 6, 7, 8, 9, 11, 12, and 14). Of the 34 transitions, 12 transitions included only a replay of the video, 9 transitions included an additional verbal prompt, 8 transitions included both verbal and gestural prompts, and 5 transitions included verbal and physical prompts. When the video-modeling intervention was withdrawn, Jose's independent transitions decreased to a mean of 27% (range = 10% to 40%). However, when the handheld video-modeling intervention was reimplemented, Jose's independent transitions from place to place increased to a mean of 82% (range = 60% to 100%). Video clips were replayed during a total of 16 transitions (i.e., Sessions 21, 22, 23, 24, and 26). Of the 16 transitions, 8 transitions included only a replay of the video, 5 transitions included an additional verbal prompt, and 3 transitions included both verbal and gestural prompts. No physical assistance was needed. Jose maintained a level of 100% independent transitions 9 weeks later.

Ida demonstrated a mean of 8% independent transitions from place to place during baseline. During the handheld video-modeling phase, Ida increased independent transitions to a mean of 79% (range = 50% to 100%). Video clips were replayed during a total of 21 transitions (i.e., Sessions 6, 7, 8, 9, 10, 11, and 12). Of the 21 transitions, 12 transitions included only a replay of the video model, 7 transitions included an additional verbal prompt, and 2 transitions included both verbal and gestural prompts. When the video-modeling intervention was withdrawn, Ida's independent transitions decreased to a mean of 47% (range = 40% to 50%). However, when the handheld video-modeling intervention was reimplemented, Ida's independent transitions from place to place increased to a mean of 88% (range = 50% to 100%). Video clips were replayed during a total of 5 transitions (i.e., Session 19) and no additional prompts were needed. Ida maintained a level of 100% independent transitions 9 weeks later.

Dave demonstrated a mean of 10% independent transitions from place to place during baseline. During the handheld video-modeling phase, Dave increased independent transitions to a mean of 74% (range = 10% to 100%). Video clips were replayed during a total of 39 transitions (i.e., Sessions 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, and 17). Of the 39 transitions, 16 transitions included only a replay of the video model, 16 transitions included an additional verbal prompt, 6 transitions included both verbal and gestural prompts, and 1 transition included verbal and physical prompts. When the video-modeling intervention was withdrawn, Dave's independent transitions decreased to a mean of 47% (range = 30% to 50%). However, when the handheld video-modeling intervention was reimplemented, Dave's independent transitions from place to place increased to a mean of 90% (range = 70% to 100%). Video clips were replayed during a total of 10 transitions (i.e., Sessions 24, 26, 29, and 30). Of the 10 transitions, 8 transitions included only a replay of the video, and 2 transitions included an additional verbal prompt. No physical assistance was needed. Dave maintained a level of 90% independent transitions 9 weeks later; however, Dave transitioned after the watching the video again and did not require additional assistance.

Social Validity

On the basis of the IRP-15 (Martens et al., 1985), the four general education teachers indicated that using video-modeling procedures by means of a video iPod in a general education classroom to increase independent transitions from place to place for elementary students with autism was socially acceptable. The teachers' mean IRP-15 score was 84 (range = 83 to 87). All teachers indicated that they agreed or strongly agreed across all

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items, specifically, (a) the intervention was "an acceptable intervention for the student's problem behavior," (b) "most teachers would find the intervention appropriate," (c) "I would suggest this intervention to other teachers," (d) "this intervention would be appropriate for a variety of students," and (e) "this intervention was beneficial to the student." The teachers also indicated that they liked the portability of the handheld device, because the students were in many different settings and have multiple transitions throughout the school day.

All the students responded similarly to the social validity semistructured interview. The students indicated that they liked to watch the videos and that it was fun watching themselves. They also suggested that they would like to watch more videos at school and implied that they got into less trouble.

Discussion

The purpose of this study was to examine the effects of using video modeling via a video iPod and response-based prompting procedures in general education classrooms or settings for elementary students with ASD, who often display inappropriate behaviors during transitional situations. Prior to the study, students required consistent teacher assistance to transition successfully from place to place. In all instances, video modeling and the use of the video iPod resulted in independent transitions, although additional prompts were required at times. Moreover, student inappropriate behaviors decreased to zero levels during transitional situations. Therefore, the video-modeling and response-based prompting intervention procedures resulted in the establishment of a functional relation, because experimental control occurred by demonstration of a covariation between change in behavior patterns and introduction of the intervention within, at least, three different series at three different points in time (Horner et al., 2005).

Consistent with prior video-modeling research, this study successfully taught students with autism a new skill (e.g., Cihak et al., 2006; Haring et al.,1987; Nikopoulos & Keenan, 2004; Sherer et al., 2001; Taylor et al.,1999) and improved behavioral functioning (Buggey, 2005; Coyle & Cole, 2004). In addition, this study supports previous video-modeling studies that demonstrated sustained positive gains across time (Buggey, 2005; Schreibman et al., 2000).

This study also extended the video-modeling research literature in several ways. First, the video-modeling intervention was used in the general education classroom in a public school setting. Educational experiences in the general education classroom provide students with disabilities the opportunity to learn and to practice skills they need to fully participate in inclusive environments. Positive gains in language development and social competency occurred when students with severe disabilities were included in general education classrooms (Rafferty et al., 2003). However, Mesibov and Shea (1996) suggested that placement in a general education classroom can be counterproductive for students with autism when faced with quickly changing schedules. Students with autism might experience difficulty following the flow of activities and keeping up with peers and ironically become dependent on adults in inclusive classes. The degree to which students with disabilities participate in the general education classroom is often dependent on their ability to demonstrate appropriate behaviors in the classroom. For students who exhibit inappropriate behaviors, access to the general education classroom may be limited. Moreover, inappropriate social behaviors is one of the most frequent reasons students with disabilities were educated outside the general education classroom. This study demonstrated that students with ASD who exhibited severe behavior problems can improve behavioral functioning in the general education setting by the means of video modeling.

In addition, students maintained transitional skills 9 weeks later. Greenwood, Delquadri, and Hall (1984) and Ivarie (1986) noted that repeated practice and increased accurate response trials facilitates skill development and maintenance. Because video-modeling and response-based prompting procedures continued until the maintenance probe, students had more opportunities to practice the skills and perform transitional behaviors successfully. Moreover, as students' transitional skills improved, they may have increasingly accessed naturally maintaining contingencies. Stokes and Baer (1977) suggested that perhaps a child needs only to be introduced adequately to the natural reinforcers, and then the behaviors eventually do not need to be maintained by continued contrived modification of the environment.

Second, students used a different device (i.e., iPod) to view themselves transitioning successfully from place to place. Although previous video-modeling studies successfully reduced problematic behaviors (Buggey, 2005; Schreibman et al., 2000), students were unable to access the intervention in multiple settings because the primary medium used was a television and videocassette recorder. Either the students had to return to the setting where the television and videocassette recorder were located or watch the video and possibly disrupt the learning of others in the classroom. This study demonstrated the versatility of video modeling. Because videos were displayed

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using a video iPod, students were able to use the intervention in any environment. The small size of the device, its portability, and the little amount of storage space required permitted the students to use the intervention anywhere. With that mind, students could rewatch the videos when requiring additional prompts. These advantages improve the usefulness of video modeling as a behavior change intervention.

Third, previous studies simply replayed the video when a student was observed making an error. This study incorporated the use of a response-based prompting strategy (i.e., system of least prompts) to facilitate appropriate transitional behaviors. Although having the student watch the video again resulted in error performance corrections (Le Grice & Blampied, 1994; Tiong, Blampied, & Le Grice, 1992), adding a response-based prompting strategy may be a viable solution when inappropriate behaviors persist. All students did require additional teacher assistance in conjunction with watching the video again; however, additional teacher prompts were fewer and less intrusive during the second intervention phase. Specifically, Ida did not require any additional teacher prompts. Also, Jose and Dave did not require physical assistance to transition from place to place. Moreover, no students required additional teacher prompts 9 weeks later. With that in mind, the use of video modeling and response-based strategies assisted in the improvement of students' independent transitions rather than either video modeling or the system of least prompts alone.

Fourth, social validity was examined. Previous studies reported that parents and private school teachers suggested favorable acceptance of video-modeling procedures (Buggey, 2005; Schreibman et al., 2000). This study expands the social validity of video modeling by using a standardized measure of general education teachers in a public school setting as well as by interviewing the participating students. All four participating teachers agreed that using video modeling was socially valid. The intervention had immediate and long-lasting effects on the students' behavior during transitional situations. All teachers reported that they would recommend the intervention strategy to other teachers and use the procedures again for other students, if needed. The teachers reported that it is often difficult to implement complex positive behavioral supports because they have less time for one-on-one instruction in the general education classroom; however, implementing video-modeling procedures through a portable device was more manageable and did not interfere with their responsibilities to other students in the classroom. All teachers also noted that it was very easy to use this intervention in any setting at school. Furthermore, all teachers reported that they continued to use the intervention after the conclusion of this study. In addition, all the students indicated that they liked watching themselves and that it helped them get into less trouble.

Several limitations may have affected the overall interpretations of this study, and future research is needed. First, similar to other single-subject designs, a small sample size was examined. In this study, only four students participated. With this mind, conclusions must be interpreted within the context of this study. Future research is needed to verify these results and its external validity across larger samples. Second, the use of a response-based prompting strategy was incorporated with video modeling as a correction procedure, and the strategies could not be parceled out. Therefore, the improvement of students' transitions solely through the use of video-modeling procedures or least prompts cannot be concluded. Future research is needed to compare the effects video-modeling interventions absent of supplemental response-based strategies. Also, future research is needed to examine how response-based prompts may be more systematically faded. Third, a functional behavior assessment or analysis was not conducted prior to intervention implementation, so the function of students' behavior was not determined. Video-modeling research is needed to examine the impact of this intervention on specific functions of behavior (e.g., escape, attention, sensory, multiple functions). Fourth, all students were elementary age and had a diagnosis of ASD. Future video-modeling research is needed to examine its effects across students with varying abilities, ages, and grade levels. Fifth, all students were highly motivated and enjoyed using the video iPod device, which may have produced intervention novelty effects. Evaluations across longer periods of time are required. Nevertheless, all students acquired the skills necessary to transition independently to different environments, and teachers reported favorable opinions concerning the video-modeling procedure and outcomes to improve behavioral functioning.

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